Roll No. | | | | | | | |

Mid-term Semester Examination 2025

Name of the Program: M. Tech(VLSI Design and Systems) *Semester: IIIrd*
Name of the Course: Organic Electronic Devices and Circuits *Course Code:* ***VDM 341***

Time: 1-1/2 Hour Maximum Marks: 50

Note:-

(i) Answer all the questions by choosing any one of the sub questions.
(ii) Each question carries 10 marks

Q1	(10marks)	**CO1, CO2**
(a)	Explain the working principle of an Organic Thin Film Transistor (OTFT) along with a clear and appropriate diagram. Also, provide the output and transfer characteristic curves with corresponding diagrams.	
	or	
(b)	Describe the distinction between conductive polymers and small molecules Using a relevant example.	
Q2	(10 marks)	**CO1, CO2**
(a)	Why is the bottom gate OTFT preferred compared to the top gate OTFT? Explain the reasons and outline the fabrication process flow diagram for OTFTs with both top contact and bottom contact structures.	
	or	
(b)	Describe the conduction mechanism in conducting polymers and illustrate it with a suitable diagram.	
Q3	(10 marks)	**CO1, CO2**
(a)	Discuss the advantages and disadvantages of organic electronics compared to inorganic electronics, supported by relevant examples.	
	or	
(b)	Describe the different applications of and explain the advantages and disadvantages of OTFTs compared to conventional MOSFETs.	
Q4	(10 marks)	**CO2**
(a)	What are the key performance parameters that influence the operation of OTFTs, and how do they affect their functioning? Please explain each parameter.	
	or	
(b)	Explain the fabrication process of OTFTs with top contact and bottom contact configurations, including flow diagrams. Also, highlight how this fabrication differs from that of conventional MOSFETs.	
Q5	(10 marks)	**CO2**
(a)	Describe the various organic thin film transistor (OTFT) structures with neat diagram.	
	or	
(b)	Describe energy band diagram for Organic-TFT with neat diagram.	

Note for the question paper setters:

- **Question paper should cover all the COs of the course.**